

HDOC Day-13

NAME: Yug Nirwal
SAP ID: 590033354
SUBJECT: C programming
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Question 1: Count Even and Odd Digits

Problem: Input a positive integer and count how many of its digits are even and how many are odd.

1. Algorithm

1. Start.
2. Input a positive integer num.
3. Set even = 0 and odd = 0.
4. Repeat while num > 0:
5. Find the last digit using digit = num % 10.
6. If digit is divisible by 2, increase even by 1; otherwise increase odd by 1.
7. Remove the last digit using num = num / 10.
8. Display the number of even and odd digits.
9. Stop.

2. Prepared Test Cases

TC	Input	Expected Output
1	12045	Even digits = 3; Odd digits = 2
2	12345	Even digits = 2; Odd digits = 3
3	2468	Even digits = 4; Odd digits = 0
4	13579	Even digits = 0; Odd digits = 5

3. C Program

```
#include <stdio.h>

int main() {
    int num, digit, even = 0, odd = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &num);

    while (num > 0) {
        digit = num % 10;
```

```

        if (digit % 2 == 0)
            even++;
        else
            odd++;

        num = num / 10;
    }

    printf("Even digits = %d\n", even);
    printf("Odd digits = %d\n", odd);

    return 0;
}

```

4. Compile the Program

Command: gcc q1.c -o q1

Compilation status: Successful (no compilation errors).

5. Execute and Test the Program

Execution command: ./q1

TC	Input	Expected Output	Actual Output	Result
1	12045	Even digits = 3; Odd digits = 2	Even digits = 3; Odd digits = 2	Pass
2	12345	Even digits = 2; Odd digits = 3	Even digits = 2; Odd digits = 3	Pass
3	2468	Even digits = 4; Odd digits = 0	Even digits = 4; Odd digits = 0	Pass
4	13579	Even digits = 0; Odd digits = 5	Even digits = 0; Odd digits = 5	Pass

```

(kali㉿kali)-[~/Desktop/cprog]
$ gcc evenNodd.c -lm

(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
Enter a psoitive integer: 12045
Even digits = 3
Odd digits = 2

(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
Enter a psoitive integer: 12345
Even digits = 2
Odd digits = 3

(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
Enter a psoitive integer: 2468
Even digits = 4
Odd digits = 0

(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
Enter a psoitive integer: 13579
Even digits = 0
Odd digits = 5

```

6. Result

The program was compiled and executed successfully. The actual outputs matched the expected outputs for all prepared test cases.

Question 2: Sum of Even and Odd Digits

Problem: Input a positive integer and find the sum of its even digits and the sum of its odd digits separately.

1. Algorithm

1. Start.
2. Input a positive integer num.
3. Set sum_even = 0 and sum_odd = 0.
4. Repeat while num > 0:
5. Find the last digit using digit = num % 10.
6. If digit is even, add it to sum_even; otherwise add it to sum_odd.
7. Remove the last digit using num = num / 10.
8. Display sum_even and sum_odd.
9. Stop.

2. Prepared Test Cases

TC	Input	Expected Output
1	12345	Sum of even digits = 6; Sum of odd digits = 9
2	12045	Sum of even digits = 6; Sum of odd digits = 6
3	2468	Sum of even digits = 20; Sum of odd digits = 0
4	13579	Sum of even digits = 0; Sum of odd digits = 25

3. C Program

```
#include <stdio.h>

int main() {
    int num, digit, sum_even = 0, sum_odd = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &num);

    while (num > 0) {
        digit = num % 10;

        if (digit % 2 == 0)
            sum_even = sum_even + digit;
        else
            sum_odd = sum_odd + digit;

        num = num / 10;
    }

    printf("Sum of even digits = %d; Sum of odd digits = %d\n", sum_even, sum_odd);
}
```

```

        num = num / 10;
    }

    printf("Sum of even digits = %d\n", sum_even);
    printf("Sum of odd digits = %d\n", sum_odd);

    return 0;
}

```

4. Compile the Program

Command: gcc q2.c -o q2

Compilation status: Successful (no compilation errors).

5. Execute and Test the Program

Execution command: ./q2

TC	Input	Expected Output	Actual Output	Result
1	12345	Sum of even digits = 6; Sum of odd digits = 9	Sum of even digits = 6; Sum of odd digits = 9	Pass
2	12045	Sum of even digits = 6; Sum of odd digits = 6	Sum of even digits = 6; Sum of odd digits = 6	Pass
3	2468	Sum of even digits = 20; Sum of odd digits = 0	Sum of even digits = 20; Sum of odd digits = 0	Pass
4	13579	Sum of even digits = 0; Sum of odd digits = 25	Sum of even digits = 0; Sum of odd digits = 25	Pass

```
(kali㉿kali)-[~/Desktop/cprog]
```

```
$ gcc sumevenNodd.c -lm
```

```
(kali㉿kali)-[~/Desktop/cprog]
```

```
$ ./a.out
```

```
Enter a positive integer: 12345
```

```
Sum of even digits = 6
```

```
Sum of odd digits = 9
```

```
(kali㉿kali)-[~/Desktop/cprog]
```

```
$ ./a.out
```

```
Enter a positive integer: 12045
```

```
Sum of even digits = 6
```

```
Sum of odd digits = 6
```

```
(kali㉿kali)-[~/Desktop/cprog]
```

```
$ ./a.out
```

```
Enter a positive integer: 2468
```

```
Sum of even digits = 20
```

```
Sum of odd digits = 0
```

```
(kali㉿kali)-[~/Desktop/cprog]
```

```
$ ./a.out
```

```
Enter a positive integer: 13579
```

```
Sum of even digits = 0
```

```
Sum of odd digits = 25
```

6. Result

The program was compiled and executed successfully. The actual outputs matched the expected outputs for all prepared test cases.